Counterfactual simulation on a budget: judging whether a cause made a difference in crowded scenes

Jadon Zhu jadonjzhu@gmail.com https://github.com/JadonJZhu/budgeted-sim

We ask whether people re-simulate only a few balls when they imagine a billiard video without the candidate cause, ball A. On the 20 videos of Lu, Bechlivanidis and Gerstenberg (2026), re-simulating one or two balls gives the lowest held-out root-mean-square error (RMSE: 18.2 rating points, against 25.1 for the counterfactual simulation model), but the gap's 95% interval includes zero. We propose 36 scenes (32 new) in which pairs of models differ by 41 to 47 rating points on average.

The question

Lu, Bechlivanidis and Gerstenberg (CogSci 2026) found that the predictions of the counterfactual simulation model (CSM; Gerstenberg et al., 2021) for whether ball A made a difference to ball B (would B have gone in without A?) drift away from people's ratings as scenes get crowded. Its predictions for how A affected B (did A change the details of B's path?) stay close. In crowded scenes people's whether and how ratings become nearly the same, even in videos built so that A changes whether B goes in but not how, or the reverse. Lu et al. read this merging as people giving up costly simulation for a contact-tracing shortcut, but do not model the shortcut; we test it as "how only" (A mattered if it reached B through a chain of contacts). Our question: when people imagine the scene without A, do they re-simulate only a few of the balls, and does a model with that limit predict their mean whether ratings better than the CSM and the shortcut?

The CSM's own noise, which perturbs only the first ball A hits, does not explain the drift. In the 9-ball whether-only video (Figure 1), A changes whether B goes in but never touches B. People's mean flipped whether rating is 20 (we flip ratings so that 100 means "B would have missed without A"), while the CSM gives B a 96% chance of missing without A, and 86% to 100% at every noise level θ the paper searched (0 to 5 degrees).

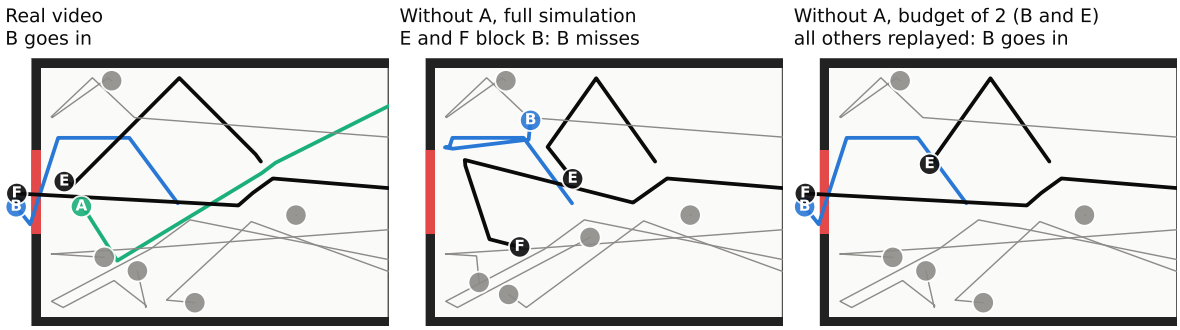


Figure 1: The 9-ball whether-only video. A sends E and F away and never touches B. Without A, a full simulation sends E and F into B, and B misses. A budget of two balls re-simulates only B and E (the first ball A hit) and replays every other ball along its real path; B then goes in, so the model says A did not matter, as people do (mean flipped rating 20).

The budget model

We propose that when people imagine the scene without A, they re-simulate only k balls, always including B, and replay every other ball along the path it took in the video. A replayed ball still pushes any re-simulated ball it meets, but is never pushed off its own path. Because replayed balls already carry A's effects, with $k = 1$, B's imagined path can change only if A hit B directly. A 1-ball budget therefore says A mattered only when that hit decided whether B went in, and A's influence through chains of other balls drops out, which is the information people seem to lose in crowded scenes. With every ball in the budget, the model is the CSM. The other $k - 1$ balls are chosen from the real video alone, before anything is imagined. We tested three rules: the balls A hit, in order of contact (lowest held-out error of the three); the balls that hit B; and the balls whose paths came closest to B's. As in the CSM, noise θ perturbs the direction of the first ball A hit. In the budget model, that ball gets the noise only if it is in the budget; otherwise no ball does. A small budget fits Balaban and Ullman (2025), who found that people update only one moving object at a time in imagination. Lu et al. cite this limit as a reason full simulation should fail in crowded scenes; we instead make the same limit the model's mechanism.

Results on the existing data

Our simulator reproduces Lu et al.'s CSM: at the paper's $\theta = 1.9$ degrees it correlates with mean whether ratings at $r = .670$ (mean of 20 seeds) against .674 for their code, and its noise-free predictions match theirs in all 20 videos.

The budget model has the lowest held-out error: RMSE 18.2 rating points against 25.1 for the CSM (Table 1). A budget of 2, filled by B and the first other ball A hit, gave the best fit to the training videos in all 20 folds. On held-out error, though, a budget fixed at 1 ties it (18.2), so the existing data do not pin down k; fit to each of the 50 people, the best budget is 1 ball for 13 people, 2 for 25 and 3 for 12. By number of balls (Figure 2), the budget model has the lowest error at 2 and 9 balls, how only at 6 and 12, and the CSM at 3, and no model's error rises steadily.

Table 1: Held-out fit to mean flipped whether ratings: each model's settings and linear rating scale are chosen on 19 videos and scored on the 20th. RMSE in rating points; Pearson r. Crowding noise puts θ on every ball whose path changes, growing with the number of balls.

Model	RMSE	r
Budget, $k = 2$ (ours)	18.2	.80
Crowding noise (θ grows with balls)	22.5	.67
How only	24.3	.59
Full CSM	25.1	.55

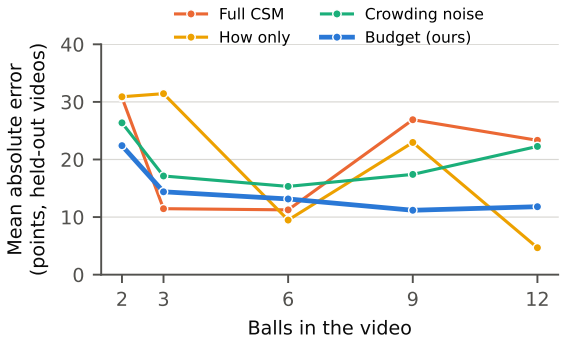


Figure 2: Mean absolute error on held-out videos, by number of balls (4 videos per point).

The budget model's lead is not decisive: in 1,000 resamples of participants, its RMSE is lower than the CSM's in 96% (by 5.4 points on average; 95% interval 10.8 lower to 0.6 higher), than crowding noise's in 81% and than how only's in 87%. No gap clears the bar we set before fitting (a 95% interval excluding zero), so the 20 videos cannot separate the models.

A test that separates the models

New scenes can separate the models where the 20 videos cannot. In the three kinds of scene built to compare two models (first three rows), the pair differs by 41 to 47 rating points on average, against at most ± 12 points (95% interval) of noise in a video's mean rating in the original data.

Kind of scene	n	Budget	Full CSM	Crowding noise	How only
Clear the way (budget vs. CSM)	8	25.2	72.5	65.8	55.5
Budget vs. crowding noise	8	25.2	65.5	68.7	50.3
Hit but no difference (budget vs. how only)	8	25.2	15.6	15.6	65.8
Budget of 1 vs. 2, and 2 vs. 3 (two scenes each way)	8	25.2 or 82.8	47.6	56.4	65.8
Original videos as anchors	4	54.0	62.0	41.2	45.2

Mean predicted flipped whether rating (0 to 100) of each model at its best fit to the existing data.

Fixed rules, not hand-picking, chose the scenes. We generated 500,000 random scenes in the style of the originals (3 to 9 balls) and kept the 37,167 that meet the original videos' constraints: B goes in, 7 to 13 seconds, every collision on screen, and no two collisions close in time and place. A scene qualifies when two models' (or two budgets') predicted probabilities differ by at least 40 points at every setting we treat as plausible: budgets of 1 to 3 balls, and every noise level whose fit is within the best fit's 95% interval.

Simple visual cues do not give the answers away, and simulated studies recover the true model. Across the 32 new scenes, no cue (A's closest approach to B, the number of collisions or balls, A passing near B's path) correlates with the gap between the budget model and any rival, or between budget sizes, above $|r| = .3$ (largest .26). In 200 simulated studies per model, that model (the generating model) produced the ratings of 50 simulated people, each copying one real participant's slider use and noise, and we refit all models blind. The generating model always fit best, and a budget model's studies were always fit best by its own k. A mix of budgets in the per-person proportions was still fit best by the budget model in all 200, with $k = 2$ in 199. These studies assume each person follows one model, so real data will separate the models less cleanly.

We propose running the 36 scenes as in Lu et al.'s Experiment 1 (about 27 minutes per participant). The budget model passes if its held-out error beats every rival's with a 95% interval excluding zero, fails if any rival beats it that way, and is otherwise inconclusive. The videos, predictions and analysis plan are in our repository, ready for your lab to run; we welcome changes.

References

Balaban, H., & Ullman, T. D. (2025). The capacity limits of moving objects in the imagination. *Nature Communications*, 16, 5899. Gerstenberg, T., Goodman, N. D., Lagnado, D. A., & Tenenbaum, J. B. (2021). A counterfactual simulation model of causal judgments for physical events. *Psychological Review*, 128(5). Lu, X., Bechlivanidis, C., & Gerstenberg, T. (2026). Causal judgments in complex situations. *CogSci 2026*. Materials: `https://github.com/cicl-stanford/complex_causation_cogsci2026`.